Roll no. : 25P-0014

Labtask 1

sks > labtask1_Q1.cpp > ...

```cpp
// Labtask 1
// Q1

#include<iostream>
using namespace std;

void swap(int*pa, int*pb);
void range(int*pa,int*pb,int*p_range);
void mean(int*pa, int*pb ,int*p_mean);
void normalized(int*pa, int*pb, int*p_na ,int*p_nb);

int main()
{
    int a,b;
    cout<<"Enter number 1 : ";
    cin>>a;
    cout<<"Enter number 2 : ";
    cin>>b;
    cout<<endl;

    int*ptra = &a;
    int*ptrb = &b;

    // initializing mean, range, normalized a and b with zero.
    int mean_var = 0;
    int range_var = 0;
    int na =0, nb =0;
```

lab_tasks > labtask1_Q1.cpp > main()

```cpp
int main()

    // pointers for mean, range, normalized a and b.
    int *p_mean = &mean_var;
    int *p_range = &range_var;
    int *p_na = &na;
    int *p_nb = &nb;

    // swap function call
    swap(ptra,ptrb);
    cout<<"******After swapping****** \n";
    cout<<"num 1 : "<<a<<endl;
    cout<<"num 2 : "<<b<<endl;
    cout<<endl;

    // range func call
    range(ptra,ptrb,&range_var);
    cout<<"Range : "<<range_var<<endl;
    cout<<endl;

    // mean function call
    mean(ptra,ptrb,&mean_var);
    cout<<"Mean : "<<mean_var<<endl;
    cout<<endl;

    // normalized function call
    normalized(ptra,ptrb,p_na,p_nb);
    cout<<"normalized values "<<endl;
    cout<<"num 1 :"<<na<<endl;
    cout<<"num 2 :"<<nb<<endl;
```

```
lab_tasks > labtask1_Q1.cpp > normalized(int *, int *, int *, int *)
int main()
    cout<< num 2 : <<nb<<endl;

    return 0;
}

void swap(int*pa, int*pb)   //akes the address of a and b as arguments
{
    *pa = (*pa)*(*pb);
    *pb = (*pa)/(*pb);
    *pa = (*pa)/(*pb);
}

void range(int*pa,int*pb,int*p_range) //takes the address of a, b and range as arguments
{
    int max =(*pa > *pb)? *pa : *pb;
    int min =(*pa < *pb)? *pa : *pb;
    *p_range = (max)-(min);
}

void mean(int*pa, int*pb ,int*p_mean) //takes the address of a, b and range as arguments
{
    *p_mean = (*pa+*pb) / 2;
}

void normalized(int*pa, int*pb, int*p_na, int*p_nb) //takes the address of a, b, normalized a and b  as arguments
{
    int na , nb;
    *p_na = (*pa)/(*pa);
    *p_nb = (*pb)/(*pb);
}
```

Output:

```
syeda-areej-e-fatima@areejfat
+ labtask1_Q1.cpp -o labtask1
num 1 : 7
num 2 : 5

Range : 2

Mean : 6

normalized values
num 1 :1
num 2 :1
```

b_tasks > labtask1_Q2.cpp > ...

```cpp
// Labtask 1
// Q2

#include<iostream>
using namespace std;

// Function to calculate the total obtained marks
void total_func(int N, int*arrA, int *p_obtained)//takes N, addres of arr and obtained marks
{
    for(int i=0; i<N; i++)
    {
        *p_obtained += *arrA;
        arrA++;
    }
}

// Function to calculate percentage
void per_func(int total, int obtained , float* p_per)// takes total and obtained marks and address of percentage
{
    *p_per = ((float)obtained*100)/(total);
}

int main()
{
    int N;
    cout<<"Enter number of subjects : ";
    cin>>N;

    int arr_o_marks[N];   // array for obtained marks
    int arr_t_marks[N];   // array for total marks
```

lab_tasks > labtask1_Q2.cpp > main()

```cpp

    int arr_o_marks[N];   // array for obtained marks
    int arr_t_marks[N];   // array for total marks

    // Intializing the variables
    int obtained = 0;
    int total = 0;
    float per = 1.0;

    // Array for storing total marks of each subject
    cout<<"***Enter the Total marks of each subject*** : "<<endl;
    for(int i = 0; i<N; i++)
    {
        cout<<"subject number"<<i+1<<" : ";
        cin>>arr_t_marks[i];
        total += arr_t_marks[i]; // adding the total marks of each subj
    }
    cout<<endl;

    // Array for storing Obtained marks of each subject
    cout<<"***Enter the Obtained marks of each subject*** : "<<endl;
    for(int i = 0; i<N; i++)
    {
        cout<<"subject number"<<i+1<<" : ";
        cin>>arr_o_marks[i];
    }
    cout<<endl;

```

```
    //total obtained marks Function call
    total_func(N ,arr_o_marks, &obtained);
    cout<<"Total : "<<obtained<<endl;

    // percentage function call
    per_func(total , obtained, &per);
    cout<<"percantage : "<<per<<"%"<<endl;

    return 0;
}
```

Output:

```
syeda-areej-e-fatima@areejfatima:~/Sem 2/OOPS_labtask/lab
ks/" && g++ labtask1_Q2.cpp -o labtask1_Q2 && "/home/syed
Enter number of subjects : 4
***Enter the Total marks of each subject*** :
subject number1 : 30
subject number2 : 30
subject number3 : 30
subject number4 : 30

***Enter the Obtained marks of each subject*** :
subject number1 : 28
subject number2 : 25
subject number3 : 20
subject number4 : 18

Total : 91
percantage : 75.8333%
syeda-areej-e-fatima@areejfatima:~/Sem 2/OOPS_labtask/lab
```